Cybersecurity Disclosure in U.S. Financial Reporting

For this study, we develop an end-to-end pipeline for constructing a media-based cybersecurity risk index by leveraging textual analysis and unsupervised filtering techniques applied to a corpus of news articles collected from five major newspapers: *Financial Times*, *Los Angeles Times*, *New York Times*, *USA Today*, and *Wall Street Journal*. The dataset comprises articles published over a ten-year period, from January 2014 to December 2023, providing a rich temporal horizon for capturing long-term trends and shifts in media attention. The pipeline ingests raw data from PostgreSQL tables corresponding to each news outlet, processes articles in batches, and assigns to each article a cyber risk relevance score. This score is derived from the relative frequency of a curated set of domain-specific keywords (e.g., "malware", "phishing", "data breach"), and is then normalized to a 0–10 scale to ensure comparability across articles and sources.

Textual preprocessing includes lowercasing, tokenization, removal of punctuation and stopwords, as well as exact duplication filtering, yielding a cleaned and deduplicated corpus that is stored in a dedicated relational structure. On top of this enriched dataset, we compute daily, weekly, and monthly indices by measuring the proportion of cybersecurity-relevant articles to the total volume of news published within each time window. The resulting proportion-based indicators serve as interpretable proxies for the evolution of public and media attention to cybersecurity issues. To assess its external validity, the behavior of the index was compared to public interest data from Google Trends, showing a strong alignment in the identification of major cyber attack episodes, thereby confirming the index's ability to effectively capture salient cybersecurity events as reported in publicly available news sources.

In designing the index, we have deliberately prioritized simplicity and transparency over more sophisticated linguistic or semantic models. This choice ensures that the methodology remains easily auditable and interpretable, facilitating validation and reproducibility across different use cases and datasets. While more advanced techniques such as named entity recognition, topic modeling, or transformer-based embeddings could potentially enhance precision, the current design allows for an intuitive link between raw inputs and index values, which is particularly valuable for policy and regulatory applications that demand explainability.

In the United States, financial disclosure obligations for publicly traded companies are governed by the U.S. Securities and Exchange Commission (SEC), which enforces a robust regulatory framework to ensure transparency and investor protection. Two of the most critical instruments in this framework are Form 10-K and Form 8-K. The Form 10-K is an annual comprehensive report that provides a detailed account of a company's financial performance

and operations. Its structure includes a wide range of sections, such as business descriptions, risk factors, financial data, management's discussion and analysis (MD&A), and governance practices. In particular, Part I of the form addresses the company's business and risk profile, while Part II covers financial disclosures, and Part III details governance and executive compensation. This form serves as the primary source of standardized financial information and strategic disclosures for investors and analysts [SEC, 2024a].

Conversely, Form 8-K is a current report used to disclose significant, material events that occur between the periodic filings of Forms 10-K and 10-Q. These events may include mergers and acquisitions, leadership changes, financial restatements, or cybersecurity incidents. Structurally, Form 8-K is organized into multiple sections, each addressing specific categories of events, such as "Registrant's Business and Operations" (Section 1), "Financial Information" (Section 2), and "Corporate Governance and Management" (Section 5). Importantly, Form 8-K is characterized by its timeliness: registrants must file it within four business days of determining that a reportable event has occurred, unless an exemption (e.g., national security) applies [SEC, 2024b].

While both forms contribute to corporate transparency, they serve distinct yet complementary functions. Form 10-K provides a backward-looking, annual synthesis of a firm's performance, including risk disclosures and financial statements subject to audit. In contrast, Form 8-K is reactive and event-driven, facilitating rapid dissemination of unforeseen material developments that may affect investment decisions. The integration of cybersecurity disclosure into both forms exemplifies this duality. Following regulatory updates in 2023, the SEC now mandates that companies disclose cyber risk management strategies, governance structures, and threat mitigation processes in Item 1C of Form 10-K, reflecting a structured and holistic view of cyber risk oversight. Simultaneously, firms must use Form 8-K to report "material" cybersecurity incidents under Item 1.05 within strict deadlines, with allowances for voluntary disclosures under Items 7.01 or 8.01 when materiality remains undetermined [Harvard Law School, 2025, McGladrey, 2025].

Recent analyses underscore the practical implications of these changes. A survey of cybersecurity-related Form 10-K filings by S&P 100 companies reveals diverse implementation strategies: while 66% use a two-part structure to distinguish between risk management and governance, others vary in their formatting or omit section headers entirely [Harvard Law School, 2025]. Additionally, a study of 8-K submissions from July 2023 to late 2024, encompassing over 9,700 companies, recorded 91 filings linked to 69 cyberattacks. Only 56% of these filings met the four-day disclosure threshold, with many firms opting for sequential updates or voluntary reporting channels due to uncertainty regarding materiality [McGladrey, 2025]. These findings reflect both the complexity and the adaptive nature of compliance in an evolving digital risk landscape.

References

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